

Course Code: PHY 121

Course Title: Physics I

Credits: 3

3.00 Contact Hours per Week

Level: Level-1, Term-II

Rationale of the Course:

Having successfully completed this course, students will be able to demonstrate knowledge and understanding of:

- Introduce the fundamental concepts of the structure of matter.
- The fundamental laws that govern heat and thermodynamics.
- To know the basic concepts of electricity and magnetism.

Course Description:

Structure of Matter: Classification of solids, Crystalline & non-crystalline solids, Single crystal and polycrystalline solids, Lattice & Basis, Crystal structure and crystal system, Coordination number and no. of atoms per unit cell, Atomic Packing Factor, Computation of APF for SC, BCC & FC, Miller indices, Miller indices calculation from crystal plane, Construction of crystal planes for different Miller indices, Defects in solids and its classification, Schottky & Frenkel defects, Bragg's law and its uses, Bonds in solids, Introduction to energy band, distinction between metal, insulator and semiconductor.

Heat and Thermodynamics: Heat and work, First law of thermodynamics and its applications, Kinetic theory of gases – kinetic interpretation of temperature, Specific heats of ideal gases, Equipartition of energy, Mean free path, Reversible and irreversible processes, Carnot's cycle, Carnot's theorem, Second law of thermodynamics, Entropy, Thermodynamic functions, Maxwell relations of thermodynamics.

Electricity and Magnetism: Electric charge and Coulomb's law, Electric field and electric field intensity, Concept of electric flux and the Gauss's law – some applications of Gauss's law, Electric potential, Relation between electric field and electric potential, Capacitor and capacitance, Electric current, Drift velocity and Current density, Resistivity, Magnetic field and magnetic force, Ampere's law, Biot-Savart law and their applications, Electromagnetic induction, Faraday's law of electromagnetic induction, Lenz law and conservation of energy.

Course Outcome (CO) Matrix:

		Bloom's Taxonomy	CP	CA	KP	Assessment Method
CO 1	Explain the fundamentals of engineering principles related to waves and oscillations, optics and modern physics.	C2	1		1	T, ASG, MT, F
CO 2	Applying the basics knowledge of waves and oscillations, optics and	C3	2		2	T, ASG, MT, F

	modern physics to demonstrate different physics laws.					
--	---	--	--	--	--	--

(CP- Complex Problems, CA-Complex Activities, KP-Knowledge Profile, T – Test; PR – Project; Q – Quiz; ASG – Assignment; Pr – Presentation; R - Report; F – Final Exam, MT- Mid Term Exam)

Mapping of COs and Program Outcomes (POs):

COs	POs (Appendix 1) H = High, M = Medium and L = Low											
	1	2	3	4	5	6	7	8	9	10	11	12
1	H											
2		M										

Justification for CO-PO Mapping:

Mapping	Level	Justification
CO1-PO1	High	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
CO2-PO2	Medium	Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Teaching Learning Strategy:

Teaching and Learning Activities	Engagement (Hours)
Face-to-Face Learning ➤ Lecture ➤ Practical / Tutorial / Studio ➤ Student-Centered Learning	42
Self-Directed Learning ➤ Non-face-to-face learning ➤ Revision ➤ Assessment Preparations	42
	21
	21
Formal Assessment ➤ Continuous Assessment ➤ Mid Term Exam ➤ Final Examination	2.5
	1.5
	3
	133

Teaching Methodology:

Lecture and Discussion, Co-operative and Collaborative Method, Problem Based Method

Content of the Course**Teaching Learning and Assessment Strategy Mapped with COs:**

Class		Intended Learning Outcomes	Teaching/ Learning Strategy	Assessment Strategy	CLO's
Week	No.				
1	1	Brief discussion and course details	Discussion		1, 2
	2	Energy level and energy band	Lecture, Q/A, feedback	Q/A, Assignment	
	3	Conductor, semiconductor and insulator	Lecture, Q/A, feedback	Q/A, Assignment	
2	4	Crystalline & non-crystalline solids	Lecture, Q/A, feedback	Q/A, Assignment	
	5	Crystal structure & crystal system	Lecture, Q/A, feedback	Q/A, Assignment	
	6	Co-ordination number & packing factor	Lecture, Q/A, feedback	Q/A, Assignment	
3	7	Computation of packing factor	Lecture, Q/A, feedback	Q/A, Assignment	
	8	Miller indices, Class test-01	Lecture, Q/A, feedback	Q/A, Assignment	
	9	Crystallographic plane from miller indices	Lecture, Q/A, feedback	Q/A, Assignment	
4	10	Defects in solids	Lecture, Q/A, feedback	Q/A, Assignment	
	11	Bragg's law and its application	Lecture, Q/A, feedback	Q/A, Assignment	
	12	Bonds in solids	Lecture, Q/A, feedback	Q/A, Assignment	

5	13	Solving mathematical problems (Structure of matter)	Lecture, Q/A, feedback	Q/A, Assignment
	14	Review of structure of matter, Assignment	Lecture, feedback	
	15	Heat and work, First law of thermodynamics and its applications, Class test-02	Lecture, Q/A, feedback	Q/A
6	16	Kinetic theory of gases	Lecture, Q/A, feedback	Q/A
	17	Kinetic interpretation of temperature	Lecture, Q/A, feedback	Q/A
	18	Specific heats of ideal gases	Lecture, Q/A, feedback	Q/A
7	19	Equipartition of energy and mean free path	Lecture, Q/A, feedback	Q/A
	20	Reversible and irreversible processes	Lecture, Q/A, feedback	Q/A
	21	Second law of thermodynamics	Lecture, Q/A, feedback	Q/A
Mid term				
8	22	Carnot's cycle	Lecture, Q/A, feedback	Q/A
	23	Carnot's theorem	Lecture, Q/A, feedback	Q/A
	24	Entropy	Lecture, Q/A, feedback	Q/A
9	25	Thermodynamic functions	Lecture, Q/A, feedback	Q/A
	26	Maxwell relations of thermodynamics	Lecture, Q/A, feedback	Q/A
	27	Solving mathematical problems (Heat and thermodynamics)	Lecture, Q/A, feedback	Q/A
10	28	Review of Heat and thermodynamics	Lecture, feedback	
	29	Electric charge and Coulomb's law, Class test-03	Lecture, Q/A,	Q/A

			feedback	
	30	Electric field and electric field intensity	Lecture, Q/A, feedback	Q/A
11	31	Concept of electric flux and the Gauss's law – some applications of Gauss's law	Lecture, Q/A, feedback	Q/A
	32	Electric potential & relation between electric field and electric potential	Lecture, Q/A, feedback	Q/A
	33	Capacitor and capacitance	Lecture, Q/A, feedback	Q/A
12	34	Electric current, drift velocity and current density	Lecture, Q/A, feedback	Q/A
	35	Laws of resistance and resistivity	Lecture, Q/A, feedback	Q/A
	36	Magnetic field and magnetic force, Ampere's law	Lecture, Q/A, feedback	Q/A
13	37	Biot-savart law and its applications	Lecture, Q/A, feedback	Q/A
	38	Laws of electromagnetic induction, Class test-04	Lecture, Q/A, feedback	Q/A
	39	Lenz law and conservation of energy	Lecture, Q/A, feedback	Q/A
14	40	Solving mathematical problems (Electricity and magnetism)	Lecture, Q/A, feedback	Q/A
	41	Review of Electricity and magnetism	Lecture, feedback	
	42	Mathematical problems solving full syllabus	Lecture, feedback	Q/A

Assessment Strategy

Components		Grading	CO	Bloom's Taxonomy
Continuous Assessment (50%)	Test 1-3	10%	CO1, CO2	C2, C3
	Assignment & Class Participation	10%	CO1, CO2	C2, C3
	Midterm	30%	CO1, CO2	C2, C3
Final Examination		50%	CO1, CO2	C2, C3

Total Marks	100%		
(CO = Course Outcome, C = Cognitive Domain, P = Psychomotor Domain, A = Affective Domain)			

Learning Materials:

Reference books:

1. Fundamental of Physics (10th)-Halliday, Resnick and Walker
2. Concept of Modern Physics (6th)- Arthur Beiser
3. Physics for Engineers (Part 1) - Dr. Giasuddin Ahmed
4. Physics for Engineers (Part 2) - Dr. Giasuddin Ahmed
5. B.Sc. Physics (Volume I) - C.L. Arora
6. B.Sc. Physics (Volume II) - C.L. Arora
7. Heat and Thermodynamics – BRIJ LAL and N. SUBRAHMANYAM

Signature, Course Teacher

Signature, HoD

Signature, Dean